



WATER THAT ISN'T WATER

Three real worlds. One impossible realm. Your organism.

Start with evidence. Notice what scientists know, what they infer, and what remains unknown. Then cross clearly into fiction.

1

EARTH

OBSERVED

Liquid-water oceans

Water covers most of Earth's surface. Life as we know it uses liquid water, energy and useful chemistry.

What does water let living things do?

2

EUROPA

SCIENTISTS INFER

A salty ocean under ice

Scientists think Europa hides a salty ocean beneath an icy crust. NASA's Europa Clipper is investigating whether it has conditions that could support life.

How could life sense a world above the ice?

3

TITAN

OBSERVED

Lakes without water

Titan has rivers, lakes and seas of liquid methane and ethane on its surface. They are not liquid water.

Could a different liquid change the rules of life?

NOW CROSS INTO FICTION

STELLAR REEF

In Mythical Void, cosmic energy flows like water. This realm and its organisms are invented.

CHOOSE YOUR STARTING RULE

☐

It flows around solid islands.

☐

It changes direction when observed.

☐

It carries light instead of heat.



DESIGN THE ORGANISM

BEGIN WITH THE PROBLEM - NOT AN EARTH ANIMAL

Your organism does not need a face, head, limbs, skin or one permanent body.

SURVIVAL RULES

HOW DOES IT HOLD TOGETHER?

HOW DOES IT SENSE CHANGE?

WHERE DOES ENERGY COME FROM?

HOW DOES IT MOVE OR SPREAD?

DRAW THE FORM - AND SHOW HOW IT CAN CHANGE

Use labels and arrows. The form can be temporary or distributed.

ONE TESTABLE PREDICTION

If the Stellar Reef flow changes in this way...

...then the organism will respond in this way, because...

RESTORE THE REEF

Which damaged part of the habitat should be restored first - and how does that help your organism?



ADULT FACILITATOR NOTE

A 45-60 minute paper-led design lab for ages 9-14

Aim: help young designers separate observation, scientific inference and fiction while imagining life for a radically unfamiliar environment.

RUN THE SESSION

- | | | | |
|---|--------|----------------------------|--|
| 1 | 8 min | Three strange seas | Compare Earth, Europa and Titan. Ask what is observed, inferred and unknown. |
| 2 | 7 min | Cross into fiction | State clearly that Stellar Reef and cosmic flow are invented. |
| 3 | 18 min | Design the organism | Start with survival problems, not an animal name or familiar body. |
| 4 | 10 min | Make a prediction | Use the If... then... because... sentence on page 2. |
| 5 | 7 min | Restore the reef | Choose one habitat repair and explain its effect on survival. |

KEEP SCIENCE AND FICTION CLEAR

- Say 'scientists think' for Europa's hidden ocean.
- Titan's surface lakes are methane and ethane, not water.
- Stellar Reef, cosmic flow and designed organisms are fiction.
- A prediction explains what would happen under the invented rules.

SAFETY AND PRIVACY

- No real methane, ethane or unknown chemicals.
- No child account or direct contact with the studio.
- Do not collect names, ages, faces, schools or contact details.
- Display work only with adult permission and no identifying details.

REAL-SPACE SOURCES

NASA Ocean Worlds: science.nasa.gov/solar-system/ocean-worlds/

NASA Europa - Ingredients for Life: science.nasa.gov/mission/europa-clipper/why-europa-ingredients-for-life/

NASA Titan Facts: science.nasa.gov/saturn/moons/titan/facts/

Independent Mythical Void activity. NASA does not endorse Mythical Void.